Supplement to: Characterize Before You Loop: a Zero-Shot Proposer's Template Does Not Survive the Loop's Own Conditions

Anonymous authors
Paper under double-blind review

Section references without an S prefix point into the main paper and are resolved from `main.aux`, so the paper builds first. Numbers in this supplement are the numbers in the paper and in the frozen arm reports; none is recomputed here. Facts attributed to the companion submission are cited to it (Anonymous authors, 2026).

S1 Ledger of the arms this paper reports

Every arm reported in this paper, with the section that reports it, its proposer, design, registered test and outcome. Design gives cells × samples per cell. Registered tests are coded P (prediction), F (falsifier), S (secondary prediction) and R (replication rule), each expanded in its arm's section. These are the ledger rows of the eight arms this paper reports, with the section column pointed at this paper. The arms were run once; arms P and P-D are also reported in the companion, and the other six appear there only by citation to this paper.

Table S1: The eight registered arms this paper reports.

Arm	§	Proposer(s)	Design	Registered test	Outcome
MU (conditioning)	4	weak tier	3 parents × 3 N × 15	P-MU1–P-MU4 + F-MU1	F-MU1 **triggered**: anchor dissolves under conditioning
L (iterated loop)	5	weak tier	2 N × 2 archives × 6 rounds × 5	L1–L4 + F-L1	L2 holds; L1 and L3 **not met** (L3 reversed); L4 open: 0/49 conditioned outputs clear the family
P (pinned path)	6	weak tier, bare API	15 cells × 15	P-P1–P-P4 + F-P1	validity collapses: 0/105 valid at the square cells, 4/75 held-out; every direct-emission prediction UNSCOREABLE; P-P3 holds (0 of 12 valid programs clear); same-day agent-runtime control 6/6 valid, anchor 4/6 (post hoc)
P-D (runtime component)	6	weak tier, bare API, $N = 13$	2 conditions × 15	P-PD1 vs P-PD2 vs P-PD3 + S-PD1	P-PD1 holds (closed at 27/30 rows by amendment, verdict unchangeable): thinking on 12/14 valid, template modal; system line only 0/13; extended thinking is the component
GM (gemini-flash-lite)	7	2nd vendor	7 N × 20	GM-chain bars	UNDERPOWERED (20 valid / 140)
GM2 (gemma, 4k budget)	7	2nd vendor	7 N × 20	GM-chain bars	0/140 parseable; budget confound, diagnosed by GM3

Arm	§	Proposer(s)	Design	Registered test	Outcome
GM3 (gemma, 16k budget)	7	gemma-4-26b	7 $N \times 20$	P-GM3.1–.3 + falsifier	pooled bar met (57.5%) but discriminating cells 11/43 = 26%, below it; misses are the rival construction
V (cross-vendor zero-shot)	7	13 free-tier aliases	5 $N \times 5$	floor + V1/V2	2 aliases TRANSFER, gemma-4-31b DOES-NOT-TRANSFER, V2 HOLDS

Registrations and commits. Conditioning arm (MU): `arm_mu_preregistration.txt` with `arm_mu_prompts.json`, committed at `2b7d202` before sampling; decision thresholds, power analysis, tie conventions and falsifier all registered. Second-vendor chain (GM, GM2, GM3): `arm_gm_preregistration.txt` at `37b3adb`, `arm_gm2_preregistration.txt` at `3019aab`, `arm_gm3_preregistration.txt` at `c0b5b97`; the serving-path deviation `arm_gm3_amendment1_openrouter.md` at `67cc4a1` was registered before the affected cells were sampled. Cross-vendor arm (V): `arm_v_preregistration.md` at `5444f10`, ancestry-checked by its runners before any sampling, with three amendments each committed before the sampling it governs. Loop arm (L): registered at `c9645fe` and scored once at `750bb32`. Pinned-path arm (P): registered at `aa473ad` before sampling. Thinking-budget arm (P-D): `arm_pd_preregistration.txt` at `5554cc0`, pushed before the first call.

Ledgers, scorers and reports. Conditioning arm: raw rows verbatim in `arm_mu_collect.jsonl`, scoring `arm_mu_analysis.py`, frozen output `arm_mu_results.txt`; the post hoc clearance recomputation is `arm_mu_ceiling.py`. Loop arm: `arm_l_collect.jsonl`, stepper `arm_l_step.py`, frozen report `arm_l_report.json`. Pinned-path arm: prompts and hashes `arm_p_prompts.json`, runner `arm_p_run.py`, scorer `arm_p_analysis.py`, report `arm_p_report.json`, ledger `arm_p_collect.jsonl` with 225 rows, six refused by the serving path; post hoc diagnostics `arm_p_diag_temperature.jsonl` and `arm_p_diag_runtime.jsonl`, labelled post hoc in their headers. Thinking-budget arm: builder `arm_pd_build.py`, spec `arm_pd_prompts.json`, runner `arm_pd_run.py`, scorer `arm_pd_analysis.py` importing the pinned-path scorer unmodified, ledger `arm_pd_collect.jsonl` with 27 rows plus the refusal rows kept for resume. Second-vendor chain: `arm_gm_gm3_checkpoint.jsonl` carries every row's raw API response verbatim with serving path and provider tagged, scored by `arm_gm3_analysis.py` into `arm_gm3_report.json` with both serving-path accountings. Cross-vendor arm: every invocation appended live to `arm_v_candidates_raw.jsonl`, scored by the companion paper's parse-validate-classify pipeline unchanged.

S2 Deviations from preregistration

The rows below are the companion paper's deviation table, restricted to the arms this paper reports and copied unchanged. Arms GM and GM2 ran under the chain registration with no deviation of their own and therefore have no row; the GM3 serving-path row is the only deviation in the chain.

Table S2: Deviations from preregistration, for the arms this paper reports.

Prediction	**Registered rule**	**As registered?**	**Outcome / reason**
GM3 serving path	first-party API only	amendment registered **before resumption** (`arm_gm3_amendment1_openrouter.md`, `67cc4a1`)	41/140 rows via routed path, provider logged; 59.0% / 57.5% dual accounting, no verdict change (§7)
Arm V proposer set	3 opencode aliases	expanded 3 → 13 by amendments 1–2, each registered **before the sampling it governs**	verdicts in §7; `big-pickle` disclosed as vendor-unattested in amendment 1

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Prediction	**Registered rule**	**As registered?**	**Outcome / reason**
Arm V transport repair	single pass, `max_tokens` 8192	amendment 3 registered **before resumption** (mechanical: 24,576 budget, retry, censored-row accounting)	877 attempt rows over the 325 registered slots (314 carry rows); floor evaluated per slot (§7)
GM3 rival-structure check	—	**post hoc**, disclosed (`diagnostics _gm3_rival.py`)	27/27 rival-valued discriminating-cell packings carry the rival's two-radius structure (§7); the registered signature is scoped to k^* and cannot see (k^*-1) structure
Arm L (iterated loop)	L1–L4	registered **before sampling** (`c9645fe`), scored once (`750bb32`); four per-lineage bars rather than one pooled bar per cell, not repaired after the fact; one post-sampling scorer correction, disclosed and given in full in §5, affecting only the 10^{-9} clearance test (corrections ledger item 34)	L2 **confirmed**; L1 **not met** on 13-DIVERSE, a 2–2 tie at $n = 4$ decided against the prediction by the registered tie rule; L3 **not met and reversed**, the prediction withdrawn rather than reframed; L4 open and answered — 0 of 49 valid conditioned outputs clear the family argmax (§5)
Arm P (pinned path)	P-P1–P-P4 + F-P1	registered **before sampling** (`aa473ad`); post-review motivation disclosed	225 invocations, 219 returning content and 6 refused by the serving path and recorded as such; validity collapsed to 0/105 at the square cells; direct-emission predictions **not evaluable** under the registered floor; P-P3 holds (0/12); F-P1 denominator empty, reported as such (§6)
Arm P temperature diagnostic	—	**post hoc**, not preregistered	0/6 valid at temperature 0.0, 0.5 and 1.0, which is 0/18 across the three settings (§6)
Arm P runtime control	—	**post hoc**, not preregistered	same prompt through the agent runtime, same day: 6/6 valid, anchor 4/6 (§6)
Arm P-D (runtime component)	P-PD1/P-PD2/P-PD3 + S-PD1	registered **before sampling** (`5554cc0`), after an outside review asked for a diagnostic beyond temperature; run interrupted at 27 of 30 rows (14 D1, 13 D2) when the serving path refused the last three for credit; closed there by amendment 1 (`arm_pd_ amendment1_ closed_at_27.md`), since no completion could change the verdict	P-PD1 **holds** and the last three rows could not change it: thinking on 12/14 valid, modal $T(4, 13)$; system line 0/13; S-PD1 met (§6)
Arm MU clearance reread	—	**post hoc**, not preregistered	the clearance question was never registered for arm MU; 88 valid at the primary tolerance, 18 of them above the argmax by at most 1.5×10^{-6} and every one failing the 10^{-9} gate; 63 valid at 10^{-9}, none above the argmax (§4) (`arm_mu_ceiling.py`)

S3 Prompts and hashes

A.1 Bare template (`{n}` parameterized; the $N = 13$ instance hashes to `32db485bea625ff9f39f4723ebf1a0 1f337559a9e2cf567fb486928f71f7f8df`):

```
Pack exactly {n} non-overlapping circles inside the unit square [0,1]x[0,1]
so that the sum of their radii is as large as possible.
```

```
MUST hold: exactly {n} circles; every circle fully inside the unit square
[0,1]x[0,1] (x-r >= 0, x+r <= 1, same for y); no two circles overlap
(distance between centers >= sum of the two radii; touching is allowed).
Do not write or execute code - construct the packing by reasoning alone.
Output ONLY the raw Python list of {n} [x, y, r] lists. No explanation, no
code fences, no other text.
```

A.2 Registered prompt hashes. bare N=13 `32db485b...`, N=17 `8437df75...`, N=21 `a415425b...`, N=31 `a664d003...`, N=35 `3b08c56e...`, N=37 `2c0a8819...`. Arm P registered the N=43 bare hash `1208e7d2...`, which was ledger-only before that. The second-vendor arms and the cross-vendor arm send these same prompts byte-identical, with the SHA-256 restated in the registration and recomputed at call time, aborting on mismatch. The conditioning arm's mutation template is hashed in `arm_mu_prompts.json` and the loop arm reuses it verbatim for generations 1 to 5.

A.3 Dispatch wrapper (the runtime arms only, and not part of the hashed task prompt): `Do not use any tools.  Your entire final message must be the answer and nothing else.`

A.4 Parser. Completions are parsed with `ast.literal_eval` after stripping code fences. A completion failing to yield a list of N numeric triples is recorded `literal_eval_failed` and scored invalid, with the failure mode logged. Radii must be strictly positive. Containment and pairwise non-overlap are checked at the two registered tolerances.

S4 Per-cell tables for the second-vendor arms

The chain's verdict table. The registered test column is the GM-chain registration, shared by arms GM, GM2 and GM3. Arm V's verdict labels are the registration's; both passing intervals span the 50% bar.

Vendor	Model	Instrument	Registered test	Verdict
Google	gemma-4-26b	first-party API, 7 N × 20 (41/140 rows routed under amendment; dual accounting)	pooled ≥ 30%; cell-level ≥ 5; falsifier ≥ 4 fails	pooled bar met (57.5%); discriminating cells 11/43 = 26%, **below** the bar; cell-level short (modal in 2 of the 5 scoreable cells, against a bar of 5; falsifier not triggered); all three misses are the *rival construction* (27/43)
Cohere	north-mini-code	free-tier alias, 5 N × 5	V1 ≥ 50% of valid on-prediction; V2 rival-argmax, valid at discriminating cells, ≤ 10%	**TRANSFERS**, point-estimate (8/12 = 67%; CI includes bar); V2 HOLDS (0/4)
OpenAI (open-weight)	gpt-oss-20b	free-tier alias, 5 N × 5	same	**TRANSFERS**, point-estimate (11/18 = 61%; CI includes bar); V2 HOLDS (0/9)
Google	gemma-4-31b	free-tier alias, 5 N × 5	same	**DOES-NOT-TRANSFER** (0/15); V2 HOLDS (0/3)
10 further aliases	(NVIDIA, Liquid, Poolside, inclusionAI, others; one unattributed)	free-tier	floor: ≥ 10 of 25 invocations valid	BELOW-FLOOR; failure taxonomies logged, no claim in either direction

Arm GM3 per cell. Recomputable from `arm_gm_gm3_checkpoint.jsonl` by `arm_gm3_analysis.py` with no arguments.

N	predicted (4 dp)	Gemma modal sum	mode freq	valid	on-prediction	MODE-MATCH
13	1.6250	1.7761	7/9	9	0	no (misses upward)
17	2.0518	**2.0518**	19/20	20	20	**yes**
21	2.1000	2.2589	9/18	18	8	no (misses upward)
31	2.5833	2.7485	11/16	16	3	no (misses upward)
35	2.9167	**2.9167**	15/15	15	15	**yes**
37	3.0345	—	—	2	0	UNSCOREABLE
43	3.0714	—	—	0	0	UNSCOREABLE

The two non-discriminating cells are $N = 17$ and $N = 35$, and 35 of the 46 on-prediction packings sit in them. The three scoreable discriminating cells are $N = 13$, 21 and 31, where the pooled rate is 11 of 43. All 20 two-radius signatures come from $N = 17$, the one extend cell among the matches, while $N = 35$'s 15 on-prediction truncations carry no filler to detect.

Arm V per model. The three aliases that cleared the registered floor are in the paper. The other ten finished BELOW-FLOOR and claim nothing in either direction. Their failure taxonomies are transport-dominated for the opencode aliases, and roughly half transport and half parse failure for two of them. Retries mean the ledger holds far more attempt rows than slots: gemma-4-31b's 25 slots took 190 attempt rows, 165 of them transport failures, and the floor is evaluated on the 25 slots.

S5 Per-cell tables for arms P and P-D, and the template-shape null

The rows below report the same two arms, whose rows both submissions carry, copied here so that this submission can be checked on its own (Anonymous authors, 2026). No value is recomputed.

Arm P per cell. Fifteen invocations per cell. Refused rows are serving-path refusals and are counted invalid at the cell they were issued for. Validity is at 10^{-6}; no cell has a different count at 10^{-9}.

channel	N	sampled	returned	refused	valid	note
square	13	15	15	0	0	all failures overlap
square	17	15	15	0	0	all failures overlap
square	21	15	15	0	0	all failures overlap
square	31	15	15	0	0	all failures overlap
square	35	15	15	0	0	one non-positive radius, rest overlap
square	37	15	15	0	0	one count mismatch, rest overlap
square	43	15	15	0	0	one malformed row, rest overlap
held-out	50	15	15	0	4	all four sum to 2.5, family argmax 3.5295867
held-out	58	15	15	0	0	count mismatches dominate
held-out	62	15	15	0	0	count mismatches dominate
held-out	65	15	15	0	0	count mismatches and overlaps
held-out	75	15	14	1	0	count mismatches and one refusal
code	13	15	13	2	4	below the floor of five
code	21	15	15	0	6	the one code cell above the floor
code	31	15	12	3	2	below the floor of five
total	—	225	219	6	16	0 of 105 square, 4 of 75 held-out, 12 of 45 code

No valid output at any cell clears the family argmax. The code cells were executed under a 10-second timeout, and P-P3 holds on their 12 valid programs.

Arm P-D per cell. One cell, $N = 13$, two conditions, closed at 27 of 30 rows by a registered amendment after the serving path refused the last three for credit.

condition	rows	valid at 10^{-6}	valid at 10^{-9}	on the anchor
D1 (thinking on)	14	12	11	5 of 12
D2 (system line only)	13	0	0	—

Under D1 the median row spends 3160 reasoning tokens, a descriptive count carrying no registered bar, $k = 4$ is modal at 6 of 12, and one output is the family argmax 1.7761424. Under D2 every failure is an overlap, which is arm P's collapse. The two D1 failures overlap. The runtime control for the same cell is 6 of 6 valid with the anchor $T(4, 13) = 1.625$ in 4 of 6, the other two being a 3×3 grid carrying corner fillers at 1.6144 and a centre-plus-border construction at 1.7500.

The template-shape null. A proposer choosing uniformly among the family's plausible template shapes, one branch value per grid order $k \in \{k^*{-}1, k^*, k^*{+}1\}$, lands on-prediction one time in three. The companion paper derives the null and this paper applies it post hoc, as it does there (Anonymous authors, 2026). Tails are exact binomial upper tails against that rate.

arm	alias	on-prediction / valid	exact upper tail
V	`north-mini-code`	8/12 pooled	0.019
V	`gpt-oss-20b`	11/18 pooled	0.014
V	`north-mini-code`	3/4 discriminating cells	0.111
V	`gpt-oss-20b`	5/9 discriminating cells	0.145
GM3	gemma-4-26b	11/43 discriminating cells	0.895

The two pooled rates clear the null and the three discriminating-cell rates do not.

References

Anonymous authors. Capability or optimizer? what lets an LLM proposer leave its template family on circle packing, 2026. Anonymous companion submission, under review. Withheld for double-blind review.